Department of Computer Science and Engineering
Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. CSE 'A' & 'B'
Course Code & Title: CCS342 DevOps
Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Ansible Playbooks**
- **Course Outcome:** CO4
- **Topic Learning Outcome: 4b**
- **Activity Chosen: Collaborative Learning - Learning by Doing**
- **Justification:**
  Ansible Playbooks are inherently practical and are best understood through experiential learning rather than lecture-based instruction. Using a learning-by-doing approach with collaborative exercises enables students to actively apply concepts like modules, inventory, handlers, and task execution. By working in small teams, students engage in peer learning, troubleshoot together, and build confidence in writing and running real playbooks. The structured task format—with goals, hints, and verification steps—guides learners in a meaningful and task-oriented way, mirroring real DevOps workflows. This activity not only reinforces technical skills but also promotes problem-solving, teamwork, and immediate feedback through practical execution.
- **Time Allotted for the Activity: 30 Minutes**
- **Details of the Implementation:**
  - The faculty member taught the concepts of ansible playbook to the students.
  - The **collaborative hands-on activity** was conducted for the duration of 30 minutes in the classroom to reinforce learning through practice.
  - Students were divided into **pairs (teams of two)**, and each team was given a printed **worksheet containing short playbook-based exercises**, such as installing Apache, creating files, running modules, or verifying services.
  - A time duration of 20 minutes was given to complete the worksheet and last 10 minutes for discussion.
  - Each worksheet included:
    - A **task objective**
    - A **one-line hint** (module or approach)
    - A **verification step** to test the outcome
  - Students discussed within their pair and wrote the playbooks directly on the worksheet which is shown in Fig 1.
  - The students recollect and apply the recently taught concepts like modules (apt, yum, file, uri), inventory usage, task syntax, and idempotence.

- The faculty member moved around the classroom, clarified doubts, and ensured that students used correct YAML structure and module syntax.
- Teams completed the tasks within the given time of 20 minutes' duration and **submitted the filled worksheets at the end** of the session. Sample completed worksheets are shown in Fig.2.
- After collecting the worksheet, the faculty member discussed the general points from the worksheet to the entire class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	**1**	**3**	**2**	**3**	**3**	**2**	**1**	**1**	**2**	**3**	**1**	**1**	**3**	**2**	**2**

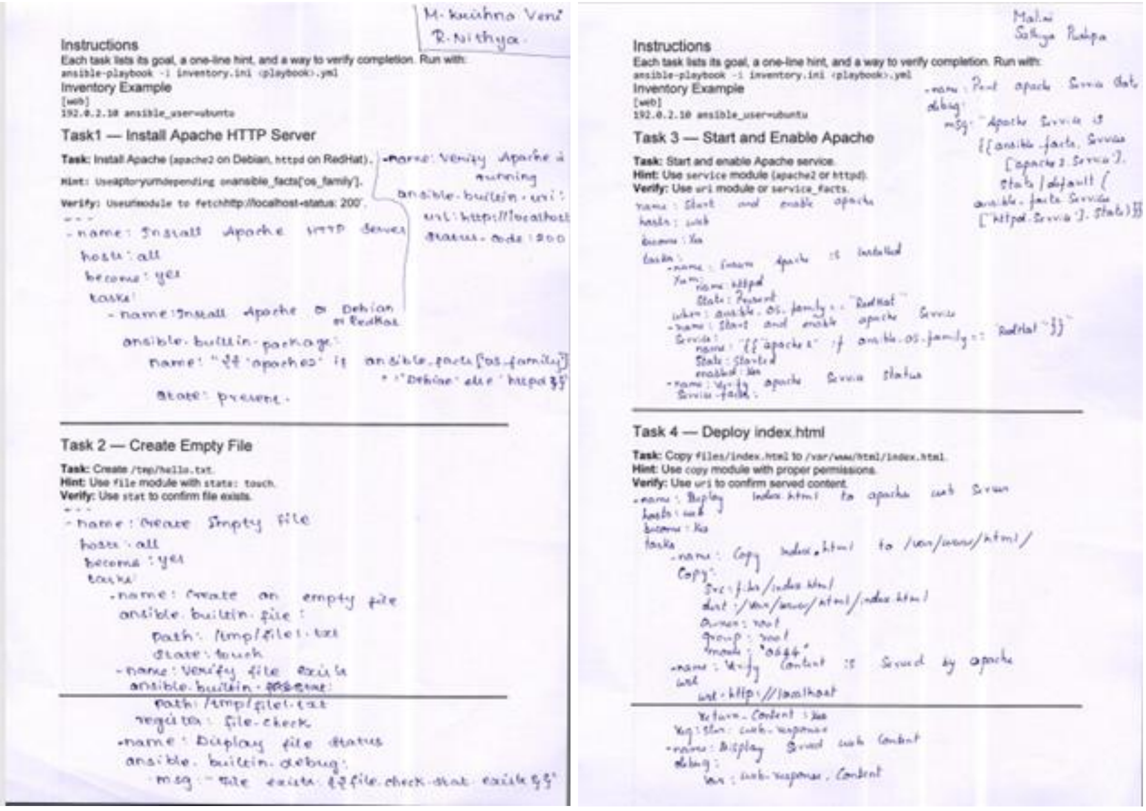
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO5**	**PO9**	**PO10**	**PSO2**
		2	**2**	**2**	**1**	**1**	**3**
Justification for correlation	Students applied foundational computing and engineering knowledge to write Ansible playbooks,	Students analyzed the tasks, identified required modules and syntax, and solved configuration problems.	Students designed small automation solutions using Ansible playbooks to install packages, manage files, or configure services	Students use Ansible, a modern automation tool, applying modules, inventory files	Teams collaborate to solve worksheet, discuss answers, and cross-verify results.	Students documented their playbook solutions clearly in the worksheet.	The concept of playbook are essential to develop cloud-aligned automation skills

- **Images / Screenshot of the practice:**

Fig. 1. Discussion Session



Instructions

Each task lists its goal, a one-line hint, and a way to verify completion. Run with:

ansible-playbook -i inventory.ini <playbook>.yml

Inventory Example

[web]

192.0.2.10 ansible_user=ubuntu

Task1 — Install Apache HTTP Server

Task: Install Apache (apache2 on Debian, httpd on RedHat).

Hint: Use apt or yum depending on ansible_facts['os_family'].

Verify: Use uri module to fetch http://localhost - status: 200.

Task 2 — Create Empty File

Task: Create /tmp/hello.txt.

Hint: Use file module with state: touch.

Verify: Use stat to confirm file exists.

Instructions

Each task lists its goal, a one-line hint, and a way to verify completion. Run with:

ansible-playbook -i inventory.ini <playbook>.yml

Inventory Example

[web]

192.0.2.10 ansible_user=ubuntu

Task 3 — Start and Enable Apache

Task: Start and enable Apache service.

Hint: Use service module (apache2 or httpd).

Verify: Use uri module or service_facts.

Task 4 — Deploy index.html

Task: Copy files/index.html to /var/www/html/index.html.

Hint: Use copy module with proper permissions.

Verify: Use uri to confirm served content.

Fig. 2. Sample Worksheet

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Students expressed that the hands-on worksheet helped them understand Ansible playbook syntax more clearly.
- Many felt that working in pairs increased confidence and peer learning.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This short, structured implementation allowed students to immediately apply concepts, clarify errors, and work collaboratively with minimal time but maximum engagement.
- Quick feedback and instructor support helped students correct mistakes instantly, leading to better learning retention.

❖ ***Challenges faced in implementation:***

- A few teams initially struggled with YAML indentation and module syntax.
- Students with lesser programming background required extra guidance during the activity.

References:

- https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning
- https://bokcenter.harvard.edu/active-learning